崑山科技大學機械工程系大學部四年制課程科目表

Kun Shan University 4-Year College Curricula (Day Division)

課程名稱(講授時數－實習時數－學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

109.11.16 系課程委員會通過
109.12.23 院課程委員會通過
110.1.27 校課程委員會通過
112.5.2 系課程委員會修訂通過
112.5.17 院課程委員會通過
112.7.20 校課程委員會通過

110 學年度入學適用

Applicable to the freshmen enrolled in Fall, 2021

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year		
第一學期 1st Semester	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	
共同必修科目(合計 30 學分) General Education Required Courses (30 Credits)								
體育分組(一) Physical Education (I) (2-0-2) 中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2) 英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	體育分組(二) Physical Education (II) (2-0-2) 勞作教育(一) Labor Education (I) (0-1-0.5) 中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2) 英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2) 校訂分類通識 School-designated Liberal Education (2-0-2)	勞作教育(二) Labor Education (II) (0-1-0.5) 校訂分類通識 School-designated Liberal Education (2-0-2) 院訂核心通識 College-designated General Core **** (2-0-2) 院訂分類通識 College-designated Liberal Education ***** (2-0-2)	服務學習與公民教育 Public Service and Civil Education (0-1-0) 校訂分類通識 School-designated Liberal Education (2-0-2)	生涯發展與職業倫理 Career Development and Work Ethics (2-0-2) 院訂分類通識 College-designated Liberal Education ***** (2-0-2)	外語選讀 Foreign Languages *** (2-0-2)			英文能力檢定 English Proficiency Examination* (1-0-1)
6-0-6	**8-1-8.5**	**6-1-6.5**	**2-1-2**	**4-0-4**	**2-0-2**			**1-0-1**
共同選修科目(合計 1 學分) General Education Selective (1 credit)								
								跨領域學習護照** Interdisciplinary Learning Passport (1-0-1)
								1-0-1

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester

註：Note:

* 「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。
"English Proficiency Examination" is enacted according to "Kun Shan University Guidelines for English Proficiency Requirement."

** 跨領域學習護照，依「崑山科技大學通識學習護照實施辦法」辦理。
"Interdisciplinary Learning Passport" is enacted according to "Kung Shan University Guidelines for General Education Learning Passport."

*** 「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。
"Foreign Languages" compromises "Foreign Languages (other than English)" and "Workplace English", among which students choose 1 course. While the courses are offered in the third academic year, "Foreign Languages (other than English)" are given preference to the sophomores.

**** 「院訂核心通識」，以開設英文相關課程為原則，課程科目表上開課時間為暫訂，課程名稱內容由各院自行規劃開設
English learning related content is given preference to offer in principle for college-designated general core courses. The schedule of this course on this curriculum chart is tentative. Course names and content are developed by the individual college.

***** 「院訂分類通識」，分別為一門數位科學類及一門院規劃通識，課程科目表上開課時間為暫訂，課程名稱內容由各院自行規劃開設
College-designated liberal education comprises one course in digital science, and one course developed by the individual colleges. The schedule of these courses on this curriculum chart are tentative. Course names and content are developed by the individual college.

院訂必修科目（合計 7 學分）General Compulsory Courses of College (Total：7 Credits)

微積分(一) Calculus Ⅰ (2-0-2) 工程技術概論 Introduction to Engineering Technology (2-0-2)		工程數學(一) Engineering Mathematics Ⅰ (3-0-3)					
4-0-4		**3-0-3**					

智慧車輛組 Intelligent Vehicles

專業必修科目(合計 69 學分) Professional Compulsory Courses (Total: 69 Credits)

普通物理及實驗(一) Physics and Experiment Ⅰ (2-1-3) 圖學 Graphics (1-2-3) @計算機程式設計 Computer Programming (1-2-3) 車輛工程 Vehicle Engineering (3-0-3)	微積分(二) Calculus Ⅱ (2-0-2) 普通物理及實驗(二) Physics and Experiment II (2-1-3) 應用力學 Applied Mechanics (3-0-3) 車輛原理及實習(一) Vehicle Principles and Practice Ⅰ (1-2-3) 電腦輔助繪圖 Computer Aided Drawing (1-2-3)	材料力學 Mechanics of Materials (3-0-3) 車輛原理及實習(二) Vehicle Principles and Practice II (1-2-3) @電子學與程式設計實務 Electric and Program Design (1-2-3)	工程數學(二) Engineering Mathematics II (3-0-3) 工程材料學 Engineering Materials (3-0-3) #電動車概論 Introduction to electric vehicles (3-0-3) 熱力學 Thermodynamics (3-0-3) 機構學 Mechanisms (3-0-3)	實務專題(一) HandsOn Ⅰ (0-1-1) 內燃機 Internal combustion engine (3-0-3) #機電儀控整合及實習 Mechatronics and Instrumentation Control and lab (1-2-3) 保養廠營運與管理 Maintenance shop operation and management （2-0-2）	實務專題(二) HandsOn II (0-1-1) #智慧車輛控制系統 Smart vehicle control system (3-0-3) 自動變速箱原理與設計 Automatic transmission principle and design (3-0-3)	暑期校外實習 Summer off-campus internship (0-*-2)	機械專業證照 Mechanical professional certificate (0-1-0) 學術倫理* Academic Ethics (1-0-1)
7-5-12	**9-5-14**	**5-4-9**	**15-0-15**	**6-3-9**	**7-1-8**	**0-*-2**	**0-1-0**

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。
The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

專業選修科目 Professional Elective Courses

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**
普通化學 General Chemistry (3-0-3) 車聯網概論 Introduction to Internet of Vehicles (2-0-2) 工業管理 Industrial Management (2-0-2)	車輛先進科技概論 Introduction to Advanced Vehicle Technology (2-0-2) 軌道車概論 Introduction to rail cars (2-0-2)	車輛專業英文 Vehicle professional English (2-0-2) 動力學 Dynamics (3-0-3) 機械加工實習(一) Machining Practice I (0-2-2) 汽車行銷學 Automotive Marketing (2-0-2)	基礎量測及實習 The Fundamental of Measurement and Experiments (1-2-3) 車輛綜合檢修 Comprehensive vehicle maintenance (1-2-3) 創意思考與方法 Creative thinking and methods (2-0-2) 模具開發實務 Mold development practice (3-0-3)	車輛動力學 vehicle dynamics (3-0-3) 流體力學 Fluid mechanics (3-0-3) 車輛機構設計 Vehicle mechanism design (2-0-2) 電動車馬達系統與實習 Electric vehicle motor system and practice (1-2-3) 自動控制 Automatic control (3-0-3) 機械元件設計 Mechanical component Design (3-0-3) 車輛感測技術 Vehicle sensing Technology （3-0-3） &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2) &機聯網證照輔導 IOT certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2)	油電混合車技術 Hybrid vehicle technology （3-0-3） 高壓電池系統與實習 High Voltage Battery Systems and Internship (1-2-3) 車用燃料學 Automotive Fuel Science (3-0-3) 車輛振動學 Vehicle Vibration （3-0-3） 車輛省能與防污技術 Vehicle energy saving and Anti-fouling technology (3-0-3) 車輛電子學及實習 Automotive Electronics and Internship (1-2-3) 燃料電池應用技術 Fuel cell application technology (3-0-3) 氣液壓技術及實習 Pneumatic and hydraulic technology and practice (3-0-3)	電動車機電整合及實習 Electric vehicle electromechanical integration and internship (1-2-3) 微處理機應用及實習 Microprocessor Application and Practice (1-2-3) 車輛 CANBUS 實務技術 Vehicle CANBUS practical technology (3-0-3) 引擎動力計及實習 Engine Dynamics and Internship (1-2-3) 電腦工程應用及實習 Computer Engineering Applications and practices (1-2-3) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internship I (0-*-9)	電動車動力系統檢修實務 Electric vehicle power system maintenance practice (1-2-3) 引擎設計 Engine design (3-0-3) 振動與噪音防治 Vibration and noise prevention (3-0-3) 可程式控制器原理與應用 PLC principles and application (1-2-3) 學年校外實習(二) Academic year off-campus internship II (0-*-9) 校外職場實習 Off-campus workplace internship (0-*-7) 高等汽車工程 Advanced Vehicle Engineering (3-0-3) 校外實習 Off-campus internship (0-*-2)

智慧製造組 Smart Manufacturing

專業必修科目(合計 73 學分) Professional Compulsory Courses (Total: 73 Credits)

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**
普通物理與實驗(一) Physics and Experiment Ⅰ (2-1-3) 圖學 Graphics (1-2-3) 機械加工實習(一) Machining Practice Ⅰ (0-3-3) @計算機程式設計 Computer Programming (1-2-3)	微積分(二) Calculus Ⅱ (2-0-2) 普通物理與實驗(二) Physics and Experiment II (1-2-3) 靜力學 Statics (2-0-2) 工具機概論 Introduction to machine tools (3-0-3) 機械加工實習(二) Machining Practice II (0-3-3)	熱力學 Thermodynamics (3-0-3) 材料學與實驗 Materials Science & Experiments (1-2-3) 2D 電腦繪圖實習 2D computer drawing practice (1-2-3) CNC 加工乙級實習(一) CNC machining Grade B Practice I （1-1-2）	機械設計 Mechanical Design (3-0-3) 3D 電腦繪圖實習 3D computer drawing practice (1-2-3) 工程數學(二) Engineering Mathematics II (3-0-3) CNC 加工乙級實習(二) CNC machining Grade B Practice II （1-1-2） @基礎量測與數位科技實務 Measurement, Digital Technology and Practice (1-2-3) 材料力學 Mechanics of Materials (3-0-3)	實務專題(一) HandsOn I (0-1-1) #進階數控加工與實習 Advanced CNC machining and practice （1-2-3） 機械製造 Mechanical Manufacture (3-0-3) #智慧製造技術 Smart manufacturing technology (3-0-3)	實務專題(二) HandsOn II (0-1-1) 自動化機械系統實習 Automated Mechanical Systems Practice （1-2-3） #工業 4.0 特色技術 Industry 4.0 Unique Technology (3-0-3)	暑期校外實習 Summer off-campus internship (0-*-2)	機械專業證照 Mechanical professional certificate (0-1-0) 學術倫理* Academic ethics (1-0-1)
4-8-12	8-5-13	6-5-11	12-5-17	7-3-10	4-3-7	0-*-2	1-1-1

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

專業選修科目 Professional Elective Courses

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**	**第一學期 1st Semester**	**第二學期 2nd Semester**
普通化學 General Chemistry (3-0-3)	動力學 Dynamics (3-0-3)	流固力實驗 Fluid Mechanics Lab (0-3-3) 氣液壓實習 Pneumatic and hydraulic practice （1-2-3） 應用性機構組立與實習 Assemble of applied Mechanism (1-2-3)	流體力學 Fluid Mechanics (3-0-3) 熱處理 Heat treatment (3-0-3) 精密軸承設計 Precision bearing design (3-0-3) 實物測繪實習 Mechanical Measurement And Drawing Practice （1-2-3）	磨潤工程 Tribology engineering (3-0-3) 節能技術導論 Introduction to Energy Saving Technology (3-0-3) 工業產品設計 Industrial product design (3-0-3) 創意思考與方法 Creative thinking and methods (2-0-2) 3D 造型設計實習 3D modeling design practice (1-2-3) 智慧製造 生產管理 Smart manufacturing production management (3-0-3)	非傳統加工 Non-traditional machining (3-0-3) 自動控制 Automatic control (3-0-3) 機械工程專題討論 Mechanical Engineering Seminar (1-0-1) 基礎科學數位教學與實習 Basic science digital teaching and practice （1-2-3） 基礎力學數位教學與實習 Basic mechanics digital teaching and practice （1-2-3） 工業管理 Industrial Management (2-0-2) &機聯網證照輔導 IOT certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2)	複合材料 Composite materials (3-0-3) 校外實習 Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習 (一) Academic year off-campus internship I (0-*-9) 模具開發實務 Mold development practice (3-0-3) &升降梯裝修證照輔導 Elevator certificate Counseling (0-*-2)	學年校外實習(二) Academic year off-campus internship II (0-*-9) &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2)

機電綠能組 Mechatronic and Green Energy Division

專業必修科目(合計 70 學分) Professional Compulsory Courses (Total: 70 Credits)

普通物理與實驗 (一) Physics and Experiment I (2-1-3) 圖學 Graphics (1-2-3) 機械加工實習(一) Machining Practice I (0-2-2) @計算機程式設計 Computer Programming (1-2-3)	微積分(二) Calculus II (2-0-2) 普通物理與實驗 (二) Physics and Experiment II (2-1-3) 機械加工實習(二) Machining Practice II (0-2-2) 2D 電腦繪圖實習 2D computer drawing practice (1-2-3)	電子學與程式設計實務 Electric and Program Design (1-2-3) 工程材料學 Engineering Materials (3-0-3) 自動控制 Automatic Control (3-0-3)	工程數學(二) Engineering Mathematics II (3-0-3) 機構學 Mechanisms (3-0-3) 3D 電腦繪圖實習 3D computer drawing practice (1-2-3) 材料力學 Mechanics of Materials (3-0-3)	PBL 專題製作 (一) Project Based Learning I (0-1-1) 熱力學 Thermodynamics (3-0-3) 應用力學 Applied Mechanics (3-0-3)	PBL 專題製作 (二) Project Based Learning II (0-1-1) 人機介面 Human Interface （1-2-3） 流體力學 Fluid mechanics (3-0-3) 工業管理 Industrial Management (2-0-2)	暑期校外實習 Summer off-campus internship (0-*-2) 熱工實驗 Thermal engineering Lab (1-2-3) #機電綠能整合與實習 Mechatronics and green energy Practice （1-2-3） #機電儀控整合實習 Mechatronics and Instrumentation Control and lab (1-2-3)	機械專業證照 Mechanical professional certificate (0-1-0) 學術倫理* Academic Ethics (1-0-1)
4-7-11	**5-5-10**	**7-2-9**	**10-2-12**	**6-1-7**	**6-3-9**	**3-*-11**	**1-1-1**

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

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專業選修科目 Professional Elective Courses

普通化學 General Chemistry (3-0-3)	綠色科技概論 Introduction to Green Science Applications (2-0-2) 綠色應用概論 Introduction to green application (2-0-2)	能源科技 Energy technology (3 -0-3)	太陽能工程概論 Introduction To Solar Energy Engineering (3-0-3) 動力學 Dynamics (3-0-3) 創意思考與方法 Creative thinking and methods (2 -0-2) 光電工程 Electro-Optic Engineering (2 -0-2)	節能技術 Energy conservation technology (3-0-3) 熱傳學 Heat transfer (3-0-3) 太陽光電技術 Solar photovoltaics technology (2 -0-2) 機械元件設計 Mechanical component Design (3-0-3) 就業技能輔導 Employment skills counseling (0-2-0) PCL 控制實習 PLC control pratice （1-2-3）	鍋爐與廢熱回收 Boiler and waste heat recovery (3 -0-3) 氣液壓技術及實習 Pneumatic and Hydraulic Engineering and Lab (3-0-3) 數值分析及實習 Numerical Methods and Lab (1 -2-3) 流固力實驗 Fluid Mechanics Lab (1 -2-3) 機械工程專題討論 Mechanical Engineering Seminar (1 -0-1)	無塵室技術 Clean room technology (2 -0-2) 有限元素法及實習 Finite element method and practice (1 -2-3) Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internshipI (0-*-9)	燃燒學 Combustion science (3 -0-3) 奈米工程概論 Introduction to nanoengineering (2 -0-2) 空調工程 Air conditioning engineering (2 -0-2) 工程論文導讀 Introduction to engineering papers (2 -0-2) 學年校外實習(二) Academic year off-campus internshipⅡ (0-*-9) 校外職場實習 Off-campus workplace internship (0-*-7)	

精密機械設計組 Division of Precision Mechanical Design

專業必修科目(合計 74 學分) Professional Compulsory Courses (Total: 74 Credits)

普通物理與實驗(一) Physics and Experiment Ⅰ (2-1-3) 圖學 Graphics (1-2-3) 機械加工實習(一) Machining Practice Ⅰ (0-2-2) @計算機程式設計 Computer Programming (1-2-3)	微積分(二) CalculusⅡ (2-0-2) 普通物理與實驗(二) Physics and Experiment II (2-1-3) 應用力學 Applied Mechanics (3-0-3) 機械加工實習(二) Machining Practice II (0-2-2) 工程材料學 Engineering Materials (3-0-3)	材料力學 Mechanics of Materials (3-0-3) 動力學 Dynamics (3-0-3) 機械製圖 Mechanical drawing (1-2-3)	工程數學(二) Engineering Mathematics II (3-0-3) 電腦輔助繪圖 Computer Aided Drawing (1-2-3) #機構學 Mechanisms (3-0-3) @基礎量測與數位科技實務 Measurement, Digital Technology and Practice (1-2-3) 機械製造 Mechanical Manufacturing (3-0-3)	實務專題(一) HandsOn I (0-1-1) 機械元件設計 Mechanical component Design (3-0-3) 熱力學 Thermodynamics (3-0-3) #電腦輔助設計及實習 Computer Aided Design & Practie (1-2-3)	實務專題(二) HandsOn II (0-1-1) 機械設計 Mechanical design (3-0-3) 流體力學 Fluid mechanics (3-0-3) 流固力實驗 Fluid Mechanics Lab (0-3-3)	暑期校外實習 Summer off-campus internship (0-*-2) #精密量測及實習 Measurement Science and practice （1-2-3）	機械專業證照 Mechanical professional certificate (0-1-0) 學術倫理* Academic ethics (1-0-1)
4-7-11	**10-3-13**	**7-2-9**	**11-4-15**	**7-3-10**	**6-4-10**	**1-2-5**	**1-1-1**

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
專業選修科目 Professional Elective Courses							
普通化學 General Chemistry (3-0-3) 綠能科技與實作 Green energy technology and practice (2-1-3)			公差原理設計 Tolerance principle design (3-0-3) 機械工程專題討論 Mechanical Engineering Seminar (1-0-1)	系統化創新設計方法 Systematic innovative design method (3-0-3) 自動控制 Automatic control (3-0-3) PLC控制實習 Programmable Logic Controller Practice (1-2-3) 物聯網與大數據於智慧製造應用 IOT and Big Data analysis applied in Intelligent Manufacturing (3-0-3) &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2)	感測器原理與應用 Sensor Principles and Applications (1-2-3) 有限元素法及實習 Finite element method and practice (1-2-3) 氣液壓技術及實習 Pneumatic and hydraulic technology and practice (3-0-3) 創意思考與方法 Creative thinking and methods (2-0-2) 機電儀控整合及實習 Mechatronic and practice (1-2-3) CPS 製造技術 CPS based Manufacturing System and Technology (3-0-3) 感測器設計 Sensor design (2-0-2) 設備減震與防噪 Equipment shock absorption and noise prevention (3-0-3)	#精密機械設計實務 Precision Machinery Design Practice (3-0-3) 熱傳學概論 Introduction to Heat Transfer (2-0-2) 數值分析及實習 Numerical analysis and practice (1-2-3) 模具開發實務 Mold development practice (3-0-3) 校外實習 Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internshipI (0-*-9)	機器手臂進階 Robot arm advancement (1-2-3) 創意性機構設計 creative mechanism design (3-0-3) 伺服系統設計 Servo system design (3-0-3) 機器動力學 Machine dynamics (2-0-2) 精密機械傳動元件技術 Precision mechanical transmission component technology (3-0-3) 高等工程分析 Advanced Engineering Analysis (3-0-3) 學年校外實習(二) Academic year off-campus internship II (0-*-9) 校外職場實習 Off-campus workplace internship (0-*-7) &機聯網證照輔導 IOT certificate Counseling (0-*-2)	

最低畢業學分：130 學分 Minimum for total credits: 130 Cs

電子與量測數位科技微學程(各組皆合計 8 學分)

智慧車輛組 Intelligent Vehicles

計算機程式設計 (1-2-3) Computer programming (1-2-3)	電子學與程式設計實務 (1-2-3) Electronics and Programming Practice (1-2-3)	通識(E 類) (2-0-2) Liberal Education(E) (2-0-2)					

智慧製造組 Smart Manufacturing

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
計算機程式設計 (1-2-3) Computer programming (1-2-3)	電子學與程式設計實務 (1-2-3) Electronics and Programming Practice (1-2-3)	通識(E 類) (2-0-2) Liberal Education(E) (2-0-2)					
機電綠能組 Mechatronic and Green Energy Division							
計算機程式設計 (1-2-3) Computer programming (1-2-3)	電子學與程式設計實務 (1-2-3) Electronics and Programming Practice (1-2-3)	通識(E 類) (2-0-2) Liberal Education(E) (2-0-2)					
精密機械設計組 **Division of Precision Mechanical Design**							
計算機程式設計 (1-2-3) Computer programming (1-2-3)	基礎量測與數位科技實務 Measurement, Digital Technology and Practice (1-2-3)	通識(E 類) (2-0-2) Liberal Education(E) (2-0-2)					

學生須修畢上述電子與量測數位科技微學程任一組之所有科目始得畢業。

Students must complete all courses within any one of the aforementioned Electronic and Measurement Digital Technology Microprogram groups in order to graduate.

電動車跨院系學分學程(合計 15 學分)智慧車輛組適用

Cross-Disciplinary Electric Vehicle Academic Program (Total: 15Credits)

電動車跨院系學分學程(合計 15 學分)智慧車輛組適用

電動車概論 (3 學分) Introduction to electric vehicles (3Credits)	智慧車輛控制系統(3 學分) Smart vehicle control system (3Credits)	機電儀控整合及實習(3 學分) 機電儀控整合及實習 Mechanical and electrical instrument control integration and practice (3Credits)	綠色能源 (3 學分) Green Energy (3Credits) (外系-學程科目) Courses from Other Departments	綠色產業 (3 學分) Green Industry (3Credits) (外系-學程科目) Courses from Other Departments

智能化製造跨院系學分學程(合計 15 學分)智慧製造組適用

Cross-Departmental Program in Intelligent Manufacturing (Total: 15Credits)

智能化製造跨院系學分學程(合計 15 學分)智慧製造組適用

智慧製造技術 (3 學分) Smart manufacturing technology (3-0-3)	工業 4.0 特色技術 (3 學分) Industry 4.0 Unique Technology (3-0-3)	進階數控加工與實習(3 學分) Advanced CNC machining and practice （1-2-3）	綠色能源 (3 學分) Green Energy (3Credits) (外系-學程科目) Courses from Other Departments	綠色產業 (3 學分) Green Industry (3Credits) (外系-學程科目) Courses from Other Departments

綠能機電整合跨院系學分學程(合計 15 學分)機電綠能組適用
Green Energy Electromechanical Integration Intercollegiate Program (Total: 15Credits)

電子學與程式設計實務(3 學分) Electric and Program Design (1-2-3)	機電綠能整合實習(3 學分) Mechatronics and green energy Practice （1-2-3）	機電儀控整合及實習(3 學分) Mechatronics and Instrumentation Control and lab (1-2-3)	綠色能源 (3 學分) Green Energy (3Credits) (外系-學程科目) Courses from Other Departments	綠色產業 (3 學分) Green Industry (3Credits) (外系-學程科目) Courses from Other Departments

機電整合跨院系學分學程(合計 15 學分)精密機械組適用
Mechatronics Interdisciplinary Program(Total: 15Credits)

機電整合跨院系學分學程(合計 15 學分)精密機械設計組適用

電腦輔助設計及實習(3 學分) Computer Aided Design & Practie (1-2-3)	精密量測及實習 (3 學分) Dimensional Metrology and Practice (3Credits)	機構學 (3 學分) Mechanisms (3Credits)	綠色能源 (3 學分) Green Energy (3Credits) (外系-學程科目) Courses from Other Departments	綠色產業 (3 學分) Green Industry (3Credits) (外系-學程科目) Courses from Other Departments

學生須修畢至少任一跨院系學程(上述機械系自訂、或工學院或他院之跨院系學分學程)始得畢業。
Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.

註：

1. 93 學年度起本系部分科目舉行共同會考，該類科目必須通過會考得以承認所修習之學分數。
   Starting from the 2004 academic year, certain courses in this department will administer a common examination. Students must pass this common examination to receive academic credit for the corresponding courses.
2. 暑期校外實習、校外實習課程皆於於暑期實施，前 160 小時為必修，可抵暑期校外實習 2 學分，第 161-320 小時為選修，可抵校外實習 2 學分；實習期滿成績及格者方給予承認課程並抵免學分。「學期校外實習」課程須於同一機構全職實習連續至少 18 週(不可分次實施) 。「學年校外實習(一)、(二)」課程須於同一機構全職實習連續至少 36 週。以上校外實習課程另訂有作業要點，修習前須依辦法提出申請。另外，學期校外實習或學年校外實習(二)可抵暑期校外實習必修 2 學分，與校外職場實習選修 7 學分。
   Both the summer off-campus internship and the off-campus internship courses are implemented during the summer vacation. The first 160 hours are compulsory and can be credited for 2 credits of the summer off-campus internship (compulsory). Hours 161-320 are elective and can be credited for 2 credits of the off-campus internship (elective). Course recognition and credit exemption will only be granted to those who successfully complete the internship and pass the assessment. The 'Semester Off-Campus Internship' course requires full-time internship at the same institution for at least 18 consecutive weeks (cannot be done in separate periods). The 'Academic Year Off-Campus Internship (I) and (II)' courses require full-time internship at the same institution for at least 36 consecutive weeks. Operational guidelines for the aforementioned off-campus internship courses

are separately stipulated, and applications must be submitted in accordance with the regulations before taking the courses. Furthermore, the semester off-campus internship or the academic year off-campus internship (II) can be credited for the 2 compulsory credits of the summer off-campus internship, and 7 elective credits of the off-campus workplace internship.

3. 有@標記之課程為教育部推動大專院校非資通訊系所之「電子與量測數位科技微學程」，需選修通識中心所開(通識 E 類課程)數位科學領域課程 2 學分及本系所開 6 學分，合計修滿 8 學分可獲頒此微學程證書。本系因應此微學程所開設之課程如下：
   a. 智慧車輛組、機電綠能組：計算機程式設計(3 學分)與電子學與程式設計實務(3 學分)。
   b. 智慧製造組、機密機械設計組：計算機程式設計(3 學分)與基礎量測與數位科技實務(3 學分)。

   Courses marked with '@' are part of the 'Electronic and Measurement Digital Technology Microprogram' promoted by the Ministry of Education for departments outside of the information and communication technology (ICT) fields. Students are required to take 2 credits of digital science domain courses (General Education Category E) offered by the Center for General Education and 6 credits offered by our department, totaling 8 credits, to be awarded this microprogram certificate. The courses offered by our department for this microprogram are as follows:

   a.Smart Vehicles Group, Mechatronic and Green Energy Division Group: Programming (3 credits) and Electronics and Lab (3 credits).

   b. Intelligent Manufacturing Group, Division of Precision Mechanical Design Group: Computer Programming (3 credits) and Measurement, Digital Technology and Practice (3 credits).

4. 有&標記的是彈性密集式證照輔導課程，上課時段將集中於考照前 1-2 個月實施密集式上課，選課前請詳閱系辦選課公告。

   Courses marked with '&' are flexible intensive license preparatory courses. Class times will be concentrated in intensive sessions conducted 1-2 months before the certification exam. Please refer to the department office's course registration announcement before registering for the course.